homework06

P185-习题 9.1	(A 类)	27-1; 29;
P213-习题 9.2		4; 6; 8; 16-1, 16-3; 19-1, 19-2; 25; 27-2;

27-1、设区域 D 是由曲线 $xy=4,\ xy=8,\ xy^3=5,\ xy^3=15$ 所围成的位于第一象限内的闭区域.

解: 依题意, 令 $u=xy, v=xy^3$, 则 $D: 4\le u\le 8, 5\le v\le 15$

$$x=\sqrt{u^3/v}, y=\sqrt{v/u}\Rightarrow dxdy=\begin{vmatrix} \frac{3}{2}\sqrt{\frac{u}{v}} & \frac{-1}{2}\sqrt{\frac{u^3}{v^3}} \\ \frac{-1}{2}\sqrt{\frac{v}{u^3}} & \frac{1}{2}\sqrt{\frac{1}{uv}} \end{vmatrix} dudv=\frac{dudv}{2v}$$

$$\Rightarrow S=\iint_D dxdy=\int_4^8 du\int_5^{15} dv\frac{1}{2v}=\frac{1}{2}\times 4\times \ln v\Big|_5^{15}=2\ln 3.$$

29、计算 $I=\iint_D \cos\left(\frac{x-y}{x+y}\right) dx\,dy$, 其中 D 是由 $x+y=1$, $x=0$ 及 $y=0$ 所围成的闭区域.

解: 依题意, 取 $u=x+y, v=x-y$, 则 $x=\frac{u+v}{2}, y=\frac{u-v}{2}$

$$D: x+y\le 1, x\ge 0, y\ge 0\Rightarrow u\le 1, u+v\ge 0, u-v\ge 0$$

$$\Rightarrow -u\le v\le u, 0\le u\le 1,\qquad dxdy=\begin{vmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{vmatrix} dudv=\frac{1}{2}dudv$$

$$\Rightarrow I=\int_0^1 du\int_{-u}^{u} dv\frac{1}{2}\cos\frac{v}{u}=\int_0^1 du\frac{u}{2}\sin\frac{v}{u}\Big|_{-u}^{u}$$

$$=\sin 1\int_0^1 u\,du=\frac{\sin 1}{2}u^2\Big|_0^1=\frac{1}{2}\sin 1$$

4、计算 $I=\iiint_{\Omega}\frac{1}{(1+x+y+z)^3}\,dx\,dy\,dz$, 其中 Ω 是由 $x=0,\ y=0,\ z=0,\ x+y+z=1$ 所围成的四面体.

解: 依题意, $\Omega:(x,y)\in D, 0\le z\le 1-x-y; D: 0\le x\le 1, 0\le y\le 1-x$

$$
\begin{aligned}
\Rightarrow I &= \int_0^1 dx\int_0^{1-x} dy\int_0^{1-x-y}\frac{dz}{(1+x+y+z)^3}\\
&= \int_0^1 dx\int_0^{1-x} dy\frac{-1/2}{(1+x+y+z)^2}\Big|_0^{1-x-y}\\
&= \int_0^1 dx\int_0^{1-x} dy\Big(\frac{1/2}{(1+x+y)^2}-\frac{1}{8}\Big) = \int_0^1 dx\Big(\frac{-y}{8}+\frac{-1/2}{1+x+y}\Big)\Big|_0^{1-x}\\
&= \int_0^1 dx\Big(\frac{x-1}{8}-\frac{1}{4}+\frac{1}{2(1+x)}\Big)\\
&= \frac{(x-1)^2}{16}-\frac{x}{4}+\frac{1}{2}\ln(1+x)\Big|_0^1 = \frac{1}{2}\ln 2-\frac{5}{16}.
\end{aligned}
$$

6、计算三重积分 $I=\iiint_{\Omega} xy^2z^3\,dx\,dy\,dz$, 其中 Ω 是由曲面 $z=xy$, 平面 $y=x$, $x=1$ 及 $z=0$ 所围成的区域.

解: 依题意, $\Omega: 0\le z\le xy, 0\le y\le x\le 1$

$$\Rightarrow (*)=\int_0^1 dx\int_0^x dy\int_0^{xy} dz xy^2z^3$$

$$=\int_0^1 dx\int_0^x dy xy^2\frac{z^4}{4}\Big|_0^{xy}=\frac{1}{4}\int_0^1 dx\int_0^x dy x^5y^6$$

$$=\frac{1}{4}\int_0^1 dx\,x^5\frac{y^7}{7}\Big|_0^x=\frac{1}{28}\int_0^1 x^{12}\,dx=\frac{1}{28}\frac{1}{13}x^{13}\Big|_0^1=\frac{1}{364}$$

8、将积分 $\int_0^1 dx \int_0^1 dy \int_0^{x^2+y^2} f(x,y,z)\,dz$ 改变积分次序，按 y,z,x 的先后次序积分.

解：依题意，

$$\Omega: 0 \le x \le 1, 0 \le y \le 1, 0 \le z \le x^2+y^2 \Rightarrow 0 \le y \le 1, 0 \le x \le 1, 0 \le z \le x^2+y^2$$

$$\Rightarrow 0 \le y \le 1, 0 \le z \le y^2, 0 \le x \le 1, z-y^2 \le x^2$$

$$\Rightarrow 0 \le y \le 1, 0 \le z \le y^2, \sqrt{z-y^2} \le x \le 1$$

$$\Rightarrow (*) = \int_0^1 dy \int_0^{y^2} dz \int_{\sqrt{z-y^2}}^1 dx\, f(x,y,z)$$

$$\Omega: 0 \le x \le 1, 0 \le z \le x^2, 0 \le y \le 1, z-x^2 \le y^2$$

$$\Rightarrow 0 \le x \le 1, 0 \le z \le x^2+1, z-x^2 \le y^2, 0 \le y, y \le 1$$

$$\Rightarrow (*) = \int_0^1 dx \int_0^{x^2} dz \int_{\sqrt{z-x^2}}^1 dy\, f(x,y,z)$$

$$+ \int_0^1 dx \int_{x^2}^{x^2+1} dz \int_0^1 dy\, f(x,y,z)$$

16-1、计算 $I=\iiint_{\Omega} yz\,dx\,dy\,dz$, 其中 Ω 是由三坐标平面及平面 $x+2y+z=2$ 所围成的闭区域;

解: 依题意,

$$\Omega: 0\le y\le 1, 0\le z\le 2-2y, 0\le x\le 2-2y-z$$

$$\begin{aligned}
\Rightarrow I &= \int_0^1 dy \int_0^{2(1-y)} dz \int_0^{2-2y-z} dx\, yz \\
&= \int_0^1 dy \int_0^{2(1-y)} dz yz(2-2y-z) \\
&= \int_0^1 dy\, y(1-y)z^2 - y\frac{z^3}{3}\Big|_0^{2-2y} = \frac{-4}{3}\int_0^1 y(y-1)^3\,dy \\
&= \frac{4}{3}\int_0^1 (y-1)^4 + (y-1)^3\,dy = \frac{4}{3}\Big(\frac{(y-1)^5}{5}+\frac{(y-1)^4}{4}\Big)\Big|_0^1 = \frac{1}{15}.
\end{aligned}$$

16-3、计算 $\iiint_\Omega y\sqrt{1-x^2}\,dx\,dy\,dz$, 其中 Ω 是由曲面 $y=-\sqrt{1-x^2-z^2}$, 圆柱面 $x^2+z^2=1$ 及平面 $y=1$ 所围成的闭区域.

解: 依题意

$$\Omega: x^2+z^2\le 1, -\sqrt{1-x^2-z^2}\le y\le 1$$

$$\Rightarrow (*)=\iint_D dxdz\int_{-\sqrt{1-x^2-z^2}}^{1} dy\, y\sqrt{1-x^2}$$

$$=\iint_D dxdz\sqrt{1-x^2}\frac{1}{2}y^2\Big|_{-\sqrt{1-x^2-z^2}}^{1}=\frac{1}{2}\iint_D (x^2+z^2)\sqrt{1-x^2}\,dxdz$$

$$=2\int_0^1 dx\int_0^{\sqrt{1-x^2}} dz\Big(x^2z+\frac{z^3}{3}\Big)\sqrt{1-x^2}=2\int_0^1 \Big(x^2(1-x^2)+\frac{1}{3}(1-x^2)^2\Big)\,dx$$

$$=2\int_0^1 \Big(\frac{2}{3}x^2(1-x^2)+\frac{1}{3}(1-x^2)\Big)\,dx=2\Big(\frac{2}{9}x^3-\frac{2}{15}x^5+\frac{1}{3}x-\frac{1}{9}x^3\Big)\Big|_0^1=\frac{28}{45}$$

19-1、计算 $\iiint_\Omega (y^2+z^2)\,dx\,dy\,dz$, 其中 Ω 是由抛物面 $x=y^2+z^2$ 与圆锥面 $x=2-\sqrt{y^2+z^2}$ 所围成的闭区域.

解: 依题意, $\Omega: r^2 \le x \le 2-r, 0 \le r \le 1, \theta \in [0,2\pi]$

$$\Rightarrow (*) = \int_0^1 dr \int_0^{2\pi} d\theta \int_{r^2}^{2-r} dx r^3 = 2\pi \int_0^1 dr\big(2-r-r^2\big)r^3$$
$$= 2\pi\Big(\frac{r^4}{2}-\frac{r^5}{5}-\frac{r^6}{6}\Big)\Big|_0^1 = \frac{4}{15}\pi.$$

19-2、计算 $\iiint_\Omega \frac{1}{\sqrt{x^2+y^2+z^2}}\,dx\,dy\,dz$, 其中 Ω 是由圆锥面 $z=\sqrt{x^2+y^2}$ 与平面 $z=1$ 所围成的闭区域.

解: 依题意, $\Omega: r\cos\varphi \le r\sin\varphi, r\cos\varphi \le 1 \Rightarrow \varphi \in [0,\frac{\pi}{4}], r \le \frac{1}{\cos\varphi}$

$$\Rightarrow (*) = \int_0^{\frac{\pi}{4}} d\varphi \int_0^{\frac{1}{\cos\varphi}} dr \int_0^{2\pi} d\theta\, r\sin\varphi$$
$$= 2\pi \int_0^{\frac{\pi}{4}} \sin\varphi \frac{1}{2} r^2 \Big|_0^{\frac{1}{\cos\varphi}} = \pi \int_0^{\frac{\pi}{4}} \frac{\sin\varphi}{\cos^2\varphi}\,d\varphi = \frac{\pi}{\cos\varphi}\Big|_0^{\frac{\pi}{4}} = \pi\big(\sqrt{2}-1\big).$$

25、计算 $I=\iiint_{\Omega}(x+y+z)^2\,dx\,dy\,dz$, 其中 Ω: $x^2+y^2+z^2\le 2az\ (a>0)$.

解: 依题意, $\Omega: r\le 2a\cos\varphi, \varphi\in[0,\frac{\pi}{2}], \theta\in[0,2\pi]$

$$\Rightarrow (*)=\iiint_{\Omega}(x^2+y^2+z^2)\,dv=\int_0^{\frac{\pi}{2}}d\varphi\int_0^{2\pi}d\theta\int_0^{2a\cos\varphi}r^4\sin\varphi\,dr$$

$$=2\pi\int_0^{\frac{\pi}{2}}d\varphi\,\sin\varphi\frac{1}{5}r^5\Big|_0^{2a\cos\varphi}=\frac{64}{5}\pi a^5\int_0^{\frac{\pi}{2}}d\varphi\sin\varphi\cos^5\varphi$$

$$=\frac{64}{5}\pi a^5\frac{-1}{6}\cos^6\varphi\Big|_0^{\frac{\pi}{2}}=\frac{32}{15}a^5\pi.$$

27-2、设 Ω 是椭球体 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\le 1, a,b,c>0$, 计算

$$\iiint_\Omega \sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}-\frac{z^2}{c^2}}\,dxdydz$$

解: 依题意, 取椭圆极坐标, 则 $\Omega: r\le 1, dxdydz = abc\,r^2\sin\varphi\,dr\,d\varphi\,d\theta$

$$\Rightarrow (*) = abc\int_0^{\pi} d\varphi\int_0^{2\pi} d\theta\int_0^1 dr\,r^2\sqrt{1-r^2}\sin\varphi$$

$$= 2\,abc\,\pi\,(-\cos\varphi)\Big|_0^{\pi}\int_0^1 r^2\sqrt{1-r^2}\,dr \qquad (r=\sin\ t)$$

$$= 4\pi\,abc\int_0^{\pi/2}\sin^2 t\cos^2 t\,dt = abc\,\pi\int_0^{\pi/2}\sin^2 2t\,dt = \frac{\pi}{2}\int_0^{\pi/2}\left(1-\cos 4t\right)dt$$

$$= abc\,\frac{\pi}{2}\left(t-\frac{\sin 4t}{4}\right)\Big|_0^{\pi/2} = \frac{\pi^2}{4}\,abc.$$